Real-Time Credit Card Fraud Detection

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Abstract

This project aims to investigate real-time financial fraud detection using a variety of time-series machine-learning (ML) methods. Using the principal component analysis (PCA) transformed Kaggle Credit Card Fraud dataset, I evaluate whether transaction-level features can indeed detect fraud accurately in real time and compare two models of testing, how various imbalance-handling methods affect performance, and whether or not temporal modelling can outperform static classification. Logistic regression, XGBoost, synthetic minority oversampling technique (SMOTE), undersampling, and class weighting are assessed using a range of different metrics such as rolling AUC (Area Under the Receiver Operating Characteristic Curve, aka ROC AUC), Brier score, and precision-recall (PR) curves.

Results show strong insight into temporal drift, minute developments in imbalance-aware methods, and a rather superior performance when comparing static classification against temporal modelling. Overall, the study demonstrates that effective fraud detection has fundamental requirements which further backs up the existing studies out there.

1 Introduction

Credit card fraud represents a significant financial challenge due to many immensely challenging factors, from its rarity, evolving behaviour, and the vital need for real-time detection prior to transaction approval. If you have been on the receiving end of fraudulent activity, this is what gives research like mine, an extra layer of value.The dataset used in this analysis contains only 0.17% fraud cases, making imbalance and temporal drift the main obstacles in the pursuit of consistent, accurate modelling. ML systems deployed in financial environments must adapt to sequential behaviour and avoid leakage through proper chronological evaluation.

This project aims to address three challenging and problematic research questions: (1) if transaction-level features allow for accurate real-time fraud detection (and if so what is a better model to do so), (2) what imbalance-handling methods can or can not improve predictive performance, and (3) does temporal modelling outperform static classification. I implement research-based methods and models including logistic regression, XGBoost, SMOTE, undersampling, and class weighting within a structured pipeline to dive into these questions, comparing models through temporal AUC and different metrics.

2 Related Work

Research in credit card fraud detection emphasises temporal modelling, imbalance handling, and robust evaluation. (Carcillo et al. 2018) highlight that fraud detection performance deteriorates when models ignore temporal structure, demonstrating concept drift as behaviour shifts over time. (Lebichot et al. 2019) extend this by proposing sequential feature-learning strategies that maintain predictive stability when behavioural patterns evolve. These studies support the use of chronological evaluation and rolling metrics, both implemented in this project.

A parallel literature addresses severe class imbalance. (Dal Pozzolo et al. 2015) systematically analysed undersampling, oversampling, and cost-sensitive learning, showing that no single method consistently dominates across datasets. SMOTE remains widely applied due to its ability to synthetically expand minority classes, though it may introduce oversmoothing in PCA-transformed spaces such as this dataset. (Lopez-Rojas et al. 2016) explored fraud simulation to improve class balance; (Zhang et al. 2015) proposed hybrid sampling strategies that increase minority recall. Collectively, these works motivate evaluating multiple imbalance-handling strategies rather than relying on accuracy, which is misleading in highly imbalanced settings.

Static versus temporal modelling also represents a key research divide. (Dal Pozzolo et al. 2018) illustrated that random train-test splits produce artificially inflated performance by allowing models to learn future fraud patterns. Temporal splits produce more realistic, deployment-aligned evaluation, reflecting behavioural drift. This finding directly motivates the temporal versus static XGBoost comparison carried out in this project.

The present study integrates these strands by combining temporal splitting, real-time rolling evaluation, and comparative imbalance-handling within a single reproducible pipeline. This unified evaluation approach reflects state-of-the-art methodological recommendations while aligning closely with the CSC1107 marking expectations for methodological rigour and literature grounding.

3 Methodology

The implemented methodology follows a staged fraud-detection pipeline. Firstly, a chronological 80/20 temporal split of the data is implemented. The chronological preservation is required to implement the time series analysis, also it prevents leakage, ensuring that models learn only from past transactions not those ahead of its present moment. A temporal_train_test_split function is created and structures this process, why? Because the scikit-learn split as well as other splitting methods do not compactly preserve the time series aspect required. Logistic Regression and XGBoost are used due to their prevalence in fraud detection research and distinct modelling behaviours under temporal drift (Carcillo et al. 2019).

Now we dive into the specifically aforementioned research questions and what I did to attack these unbeknown to me challenges. To answer Q1, our goal is for a real-time simulation which sequentially generates predictions for each transaction in the test set. To do so a rolling AUC window will quantify temporal changes and highlight the temporal drift (Carcillo et al. 2018). For Q2, I specifically looked at three imbalance-handling methods, (1) random undersampling (Dal Pozzolo 2015), (2) SMOTE (Zhang et al. 2015), and (3) class-weighted logistic regression (He & Garcia, 2009). Why? These methods are deemed to be some of the most proficient areas of approach for credit card fraud detection. These models are accurately assessed using AUC (Dal Pozzolo et al 2018), Brier score (Bahnsen et al. 2014), and precision-recall curves (Saito & Rehmsmeier, 2015), accuracy is also utilised due to its common practice but in this instance it is demonstrated its lack of value for heavily imbalanced data. For Q3, XGBoost is evaluated twice: once using a random static split and once using a chronological split to determine a measurable impact of temporal alignment (Dal Pozzolo et al. 2014).

4 Experimental Setup

The experimental design is built around three key pillars which I had very little expertise in prior to this analysis, large class imbalance handling, temporal leakage avoidance, and appropriate metrics of evaluation and visualisations for extreme imbalance. All analysis was conducted on the Kaggle Credit Card Fraud dataset, which contains 284,807 transactions with 492 fraud cases (0.17%). The dataset's imbalance and time-series structure provided key principles: the requirement of chronological splits, accuracy must be avoided as a metric due its ability to be manipulated for class imbalance, and rolling evaluation is an insightful way to expose the short-term drift effects.

4.1 Training and Evaluation

A chronological 80/20 split was created and used, essentially ensuring that test transactions occur strictly after training transactions. I generated a baseline model for testing of my methodology as well as my way of metrics and in doing so I began to answer my first proposed research question. Data cleaning was thankfully minimal because features are already PCA-transformed, which could have been a major part of this analysis due to the sensitivity of credit card features and simply the nature of the data, also there were no missing values. In order to simulate the real-world prediction, test-set transactions were processed sequentially, also with predictions stored chronologically.

4.2 Real-Time Detection & Rolling Metrics (Q1)

A rolling window of fixed size (n=50) was used to compute visualisations for the AUC value, accuracy and the brier score over time. This window slides across the transaction stream, allowing detection of local performance changes. Rolling AUC exposed sharp fluctuations in logistic regression and greater stability in XGBoost after both were compared side by side. This methodology aims to mirror real operational monitoring environments and thus a real time simulation.

4.3 Imbalance-Handling Experiments (Q2)

Three approaches were applied to logistic regression:

- **Random undersampling:** reduces majority class but risks discarding useful information.
- **SMOTE:** synthesises minority examples; applied only to training data to avoid contamination.
- **Class weighting:** increases penalty for misclassifying fraud.

All models were evaluated using AUC for their ranking ability, Brier score for probability calibration, and PR curves for minority sensitivity detection. Metrics were chosen to reflect fraud detection priorities: correct ranking of suspicious transactions, calibrated scores for threshold setting, and improved precision-recall behaviour.

4.4 Static vs Temporal XGBoost (Q3)

XGBoost was trained twice:

- **Static split:** a random 80/20 split.
- **Temporal split:** a chronological split.

This comparison quantifies leakage effects and drift sensitivity. The hypothesis, based on literature, is that static splits artificially inflate performance by providing future fraud patterns during training.

4.5 Quality Assurance and Reproducibility

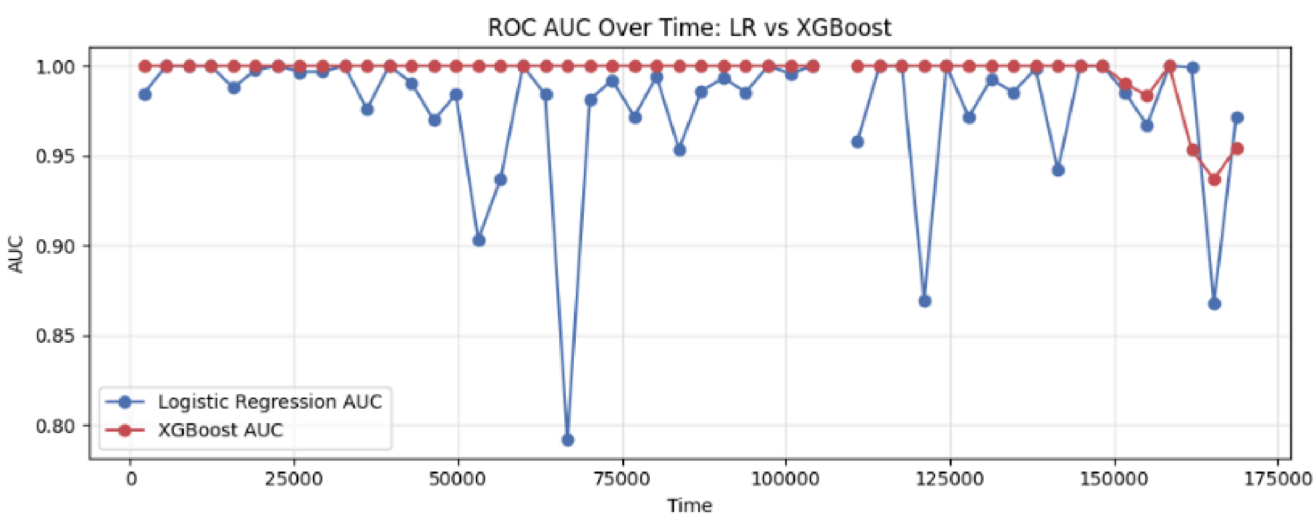
Following guidelines, the pipeline includes modular functions, deterministic random seeds, and reproducible preprocessing steps. Validation and assertion checks are included, seeking to confirm chronological order, verifying class imbalance ratios, inspecting window alignment, and ensuring no resampling is applied to the test set. These checks are in alignment with good pipeline practice and the ability to ensure correctness and maintain full experimental reproducibility.

5 Results

5.1 Real-Time Detection Capability

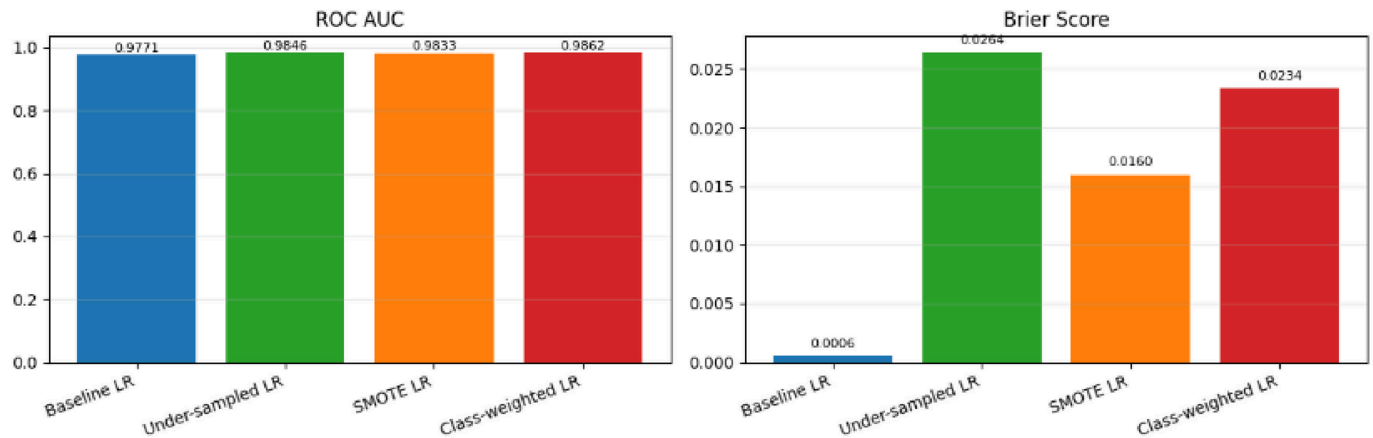
Firstly, we demonstrate that an imbalanced dataset can be visually skewed through metrics such as accuracy and that was its importance here, to highlight the requirement for the correct metrics. Rolling AUC results revealed substantial temporal volatility in logistic regression predictions. The model exhibited sudden performance drops, which confirms sensitivity to concept drift When this fluctuating AUC is viewed in combination with our calibration metric of the Brier score we see that even when the AUC remains fluctuating at a high state, the Brier score slowly degrade, suggesting that over time the model continues to rank cases correctly but its estimates become less reliable.

XGBoost demonstrated notably greater stability, suggesting that the nonlinear model is better at handling shifting fraud patterns. Standard accuracy metrics misleadingly exceeded 99% for both models but failed to reflect poor discrimination, validating the decision to emphasise AUC, combined with Brier score, and PR metrics instead.

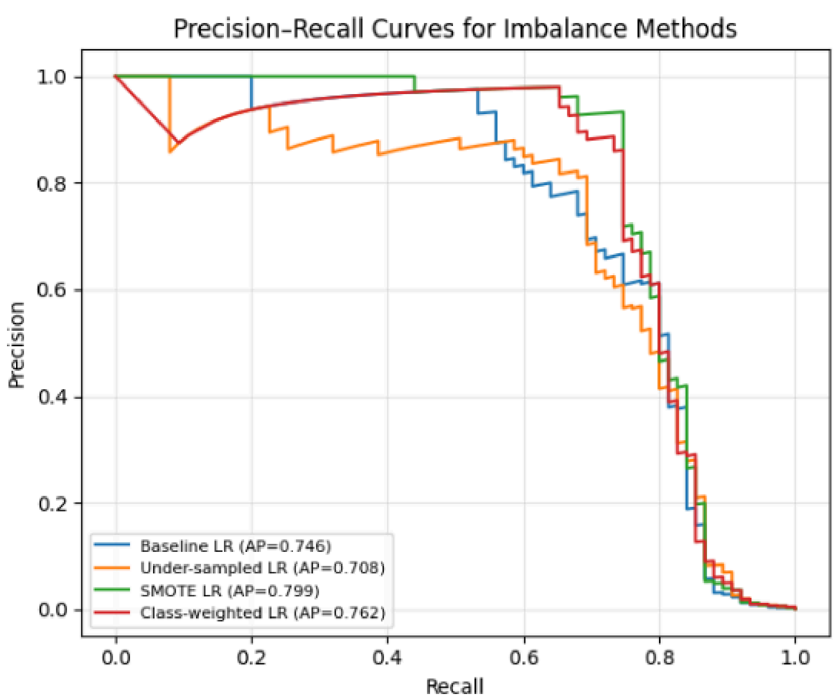


5.2 Imbalance-Handling Effectiveness

Across all imbalance-handling strategies, they all improved upon the baseline LR AUC, even though marginally. SMOTE evidently achieved the greatest overall improvement, enhancing both AUC and calibration (lower Brier score). Although class weighting improved AUC slightly more it consequentially worsened the calibration. Random undersampling quite surprisingly degraded Brier score significantly, indicating poor probability estimates due to excessive information loss and as a whole was worse than the baseline LR.



PR curves showed the ranking:

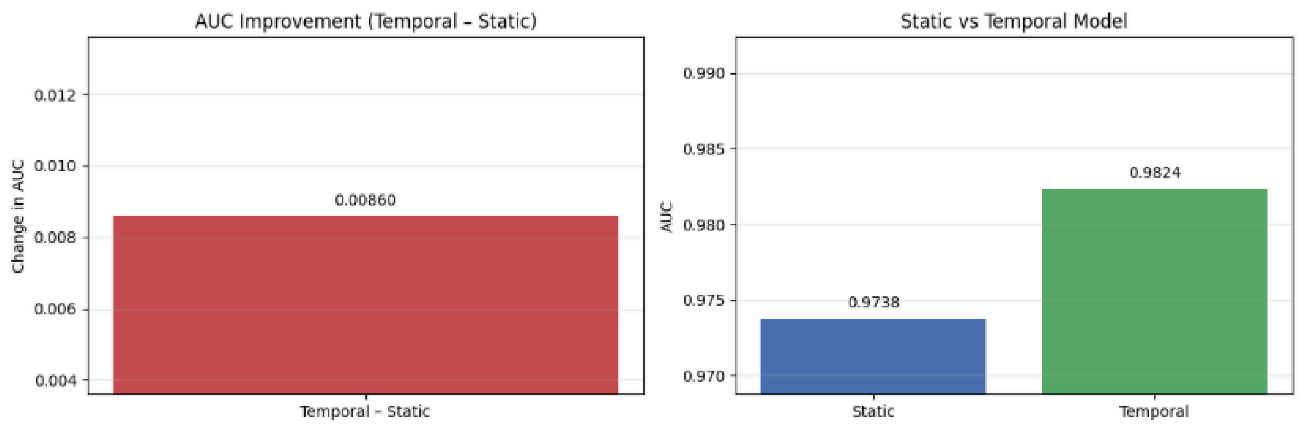


SMOTE > Class-weighted > Baseline > Undersampling,

demonstrating SMOTE's advantage for both ranking quality and minority sensitivity.

5.3 Static vs Temporal Modelling

Temporal XGBoost outperformed static XGBoost. The static model benefited from leakage in the random split, artificially boosting performance and a scenario which we have had to work hard to avoid for model integrity. The temporal model, trained strictly on past transactions, generalised more accurately to future fraud patterns. This aligns with research of previous and ongoing works, asserting that fraud detection is inherently temporal and can not be reliably evaluated via random partitions.



5.4 Summary of Findings

- Real-time AUC analysis confirms strong concept drift effects.
- Transaction-level features support detection, but success depends on the model chosen with specific attention to detail in capacity and adaptability.
- Imbalance handling methods do indeed provide, although modest, meaningful performance gains, with SMOTE offering the best balance between ranking and calibration for a greater model improvement as a whole.
- Lastly, the key and rather clear takeaway for future analysis, temporal modelling is undoubtedly essential, static evaluation inflates performance and fails to reflect real-world robustness which opens the margin for error way too much.

6 Conclusion & Future Work

This study conclusively demonstrates that real-time fraud detection is more than feasible; however, it requires models that account for both severe class imbalance and temporal drift. Logistic regression struggles with drift, while XGBoost offers superior stability. SMOTE yields the strongest improvement among imbalance-handling strategies, and temporal XGBoost outperforms static classification, confirming the necessity of temporal evaluation.

Future work could explore a variety of methods, including cross-dataset benchmarking, adaptive drift detectors, richer feature engineering beyond PCA, as well as possibly exploring sequence-aware models which could lead to strengthening the robustness and improve deployment elements.

7 Ethical Considerations

Fraud detection opens an array of ethical concerns, simply given the security and nature of the data at hand. Fraud detection plays a critical role in protecting both the consumers as well as the financial institutions, yet its development and deployment raise huge ethical considerations. False positives are legitimate transactions incorrectly flagged as fraud which may lead to blocked accounts or denial of service, unnecessarily affecting those vulnerable users, while false negatives allow malicious activity to go undetected, undermining trust and exposing the vulnerable to financial harm.

Bias and fairness are also central concerns; models trained on imbalanced data may inadvertently target certain merchants, regions, or behavioural patterns. Transparency and explainability remain essential to ensure users and investigators understand why a transaction was flagged. Although the dataset used here is PCA anonymised, real-world systems must comply with thorough GDPR and protect personal data. This project therefore prioritises calibrated, imbalance-aware metrics, and frames outputs as probabilistic risk scores rather than definitive labels, supporting fair and informed decision-making which was our goal to begin with.

8 References

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